



Amlogic Hybrid Android Q Release Note

AMLOGIC, Inc.

2518 Mission College Blvd
Santa Clara, CA 95054
U.S.A.
www.amlogic.com

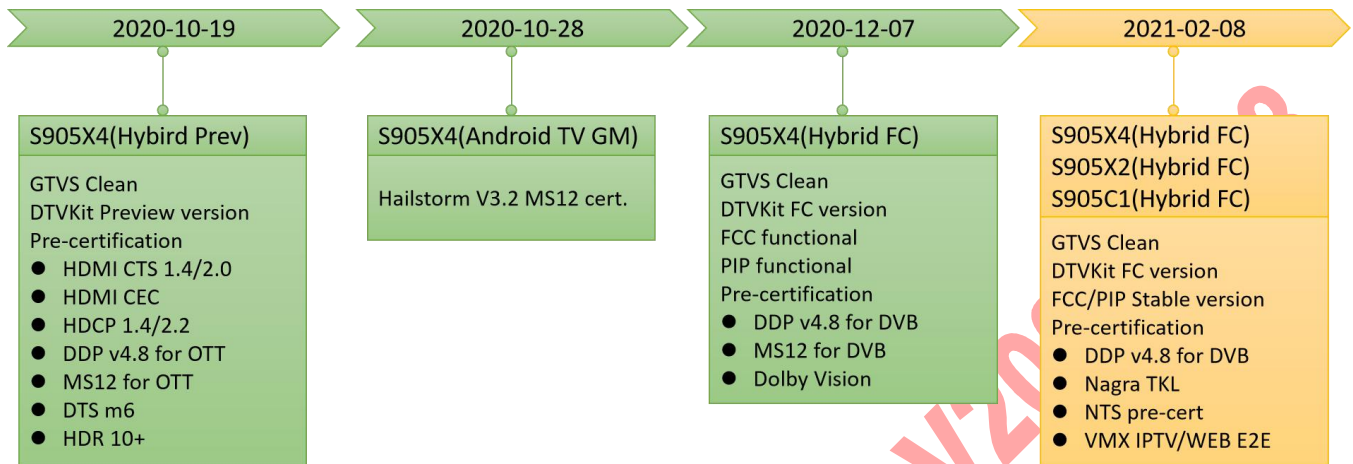
AMLOGIC reserves the right to change any information described herein at any time without notice.
AMLOGIC assumes no responsibility or liability from use of such information.

Amlogic Openlinux Release Notes

Content

1. Release Schedule.....	3
2. Basic Information.....	4
2.1. INTRODUCTION.....	4
2.2. CHIP INFORMATION.....	6
2.3. REFERENCE PLATFORM.....	7
2.4. HOW TO GET CODE.....	8
2.5. HOW TO ENABLE FCC/PIP FUNCTIONAL.....	9
2.6. HOW TO BUILD CODE.....	10
2.7. DEVICE VENDOR SCS(SECURE BOOT V3).....	12
2.8. SECURE MODE.....	13
2.9. FASTBOOT.....	14
2.9.1. adb/fastboot version.....	14
2.9.2. Two fastboot modes.....	15
2.9.3. How to remount.....	17
2.10. HOW TO UPGRADE.....	18
2.10.1. all partitions.....	18
2.10.2. physical partitions.....	19
2.10.3. logical partitions.....	20
3. Test Report.....	21
3.1. AUTO REBOOT/SHUTDOWN IN NORMAL TEMPERATURE.....	21
3.2. AUTO SUSPEND IN NORMAL TEMPERATURE.....	21
3.3. 3D GAME BURNING TEST REPORT.....	21
3.4. MONKEY TEST TEST REPORT.....	21
3.5. FACTORYRESET TEST REPORT.....	22
3.6. A/V BURN TEST REPORT.....	22
3.7. DVB BURN TEST REPORT.....	22
3.8. HIGH(70°C) TEMPERATURE TEST REPORT.....	23
3.9. LOW(-20°C) TEMPERATURE TEST REPORT.....	23
3.10. VIDEO FORMAT TEST REPORT.....	24
3.11. POWER CONSUMPTION TEST REPORT.....	25
4. Tools version.....	26
5. License version.....	26
6. Licence path.....	27
7. Attention.....	27

1. Release Schedule



2. Basic Information

2.1. Introduction

This document is the release notes of Amlogic reference source codes aligned with Android Q running on Amlogic S905X4/S905X2/S905C1 reference hardware.

S905X4/S905X2(With DTVKit middleware) and S905C1(with Irdeto) released the GTVS clean version(FC version).

The purpose of this release is:

1. GTVS Clean on S905X4(ohm).
 - (1) CTS(10.0 R6)
 - (2) GTS(8.0-r2)
 - (3) VTS(10.0 R6)
 - (4) CTS-verify(10.0 R6)
 - (5) CTS-on-gsi(10.0 R6)
 - (6) STS(2020-12)
 - (7) TVTS(1.3_R2)
 - (8) Android TV smoking(Youtube testing)
2. Pre-certification
 - (1) HDMI CTS 1.4/2.0 pass
 - (2) HDMI CEC pass
 - (3) HDCP 1.4/2.2 pass
 - (4) DTS m6 pass
 - (5) HDR 10+ pass
 - (6) Dolby vision pass
 - (7) DDP v4.8 for DVB pass
 - (8) VMX-IPTV Client E2E
 - (9) VMX-WEB Client E2E
 - (10) NTS self pre-cert pass
3. Full function ready:
 - (1) Component compatibility
 - 1) HDMI
 - 2) CVBS
 - 3) WIFI
 - 4) BT
 - 5) Ethernet(100/1000M)
 - 6) Bluetooth
 - 7) USB
 - 8) Remote
 - 9) Spdif
 - (2) PQ
 - (3) Multi-media functionality and compatibility
 - (4) Multi-Instance
 - (5) Display functionality and compatibility
 - (6) HDR10
 - (7) Dolby Vision
 - (8) Alway HDR
 - (9) System functionality and compatibility
 - (10) Audio-output
 - (11) Secure boot V3
 - (12) A/B update

Amlogic Openlinux Release Notes

- (13) Fastboot
- 4. Performance
 - (1) Benchmark
 - (2) Bootup/Suspend Speed
 - (3) UX
 - (4) APP launch speed
 - (5) Video start/Switch speed
 - (6) Memory usage
- 5. Stability.
 - (1) Auto reboot - 72h
 - (2) Auto Shutdown - 72h
 - (3) Auto suspend - 72h
 - (4) GPU - 3D Game - 72h
 - (5) Monkey test - 72h
 - (6) Factory reset - 72h
 - (7) A/V Burn Test - 72h
 - (8) High/Low Temperature Test - 72h
- 6. DTVKit
 - (1) DVB-C Program search
 - (2) DVB-S/DVB-S2 Program search
 - (3) DVB-T/DVB-T2 Program search
 - (4) Channel auto search
 - (5) Channel manual search
 - (6) NIT search
 - (7) MHEG5
 - (8) Teletext
 - (9) PVR
 - (10) TimeShift
 - (11) subtitle
 - (12) Program Edit
 - (13) LCN
 - (14) EPG
 - (15) AFD
 - (16) Channels Lock
 - (17) Parental Lock
 - (18) AD
 - (19) Info Banner
 - (20) Aspect Ratio
 - (21) update channel
 - (22) Date, Time
 - (23) Audio language
 - (24) Switch Channel
 - (25) FCC
 - (26) PIP
- 7. Nagra TKL DVT pre-cert

The known issue of this release is:

- 1. MS12 v1.3.2 for DVB has two AD case issues.
- 2. QCA 6174 TX throughput need clarify with QCA.
- 3. Few AV compatibility issues.
- 4. Few HDMI compatibility issues.
- 5. PIP/FCC have some compatibility issues.

Amlogic Openlinux Release Notes

6. Multi-AD has some issues.

2.2. Chip Information

S905X4 :

We choose S905X4/S905X2/S905C1 reference platform as our developing and testing platforms for this release.

Item	S905X4(ohm)	S905X2(franklin_hybrid)	S905C1(newton_hybrid)
CPU	Quad-core ARM Cortex-A55	Quad Cortex-A53	Quad Cortex-A55
GPU	ARM Mali-G31 MP2	ARM Mali-G31 MP2	ARM Mali-G31 MP2
Security	TrustZone & TVP	TrustZone & TVP	TrustZone & TVP
Memory	32-bit DDR3/4/3L, LPDDR3/4	32-bit DDR3/4/3L,LPDDR3/4	32-bit DDR3/3L/4, LPDDR3/4
Video decoding	AV1/H.265/VP9/AVS2-P2 to 4k@60fps	H.265/VP9/AVS2-P2 to 4k@60fps	H.265/VP9/AVS2-P2 to 4k@60fps
Video Encoding	H.265/JPEG to 1080p@60fps	H.265/JPEG to 1080p@60fps	H.265/JPEG to 1080p@60fps
HDMI-Tx	4k@60hz	4k@60hz	4k@60hz
Ethernet	10M/100M/1000M	10M/100M/1000M	10M/100M/1000M
AV output	CVBS	CVBS	CVBS
IP License	Dolby,DTS,Dolby Vision	Dolby,DTS,Dolby Vision	Dolby,DTS,Dolby Vision

2.3. Reference Platform

Mbox:

- **Ohm(S905X4)**
DDR4 2GB, EMMC, WIFI QCA6174
- **Franklin_Hybrid(S905X2)**
DDR4 2GB, EMMC, WIFI 8822cs
- **Newton_Hybrid(S905C1)**
DDR4 2GB, EMMC, WIFI 8822cs

Amlogic Openlinux Release Notes

2.4. How to Get Code

You can download Android Q source code by running the following repo commands:

1. use below command to get the code in google server and amlogic server(apply the license from Google and Amlogic, and apply the separate license for playready and tdk from Amlogic):

```
$cd ~/<your-project-repo-dir>/  
$repo init -u ssh://git@openlinux.amlogic.com/q-amlogic/platform/manifest.git -b q-amlogic -m  
openlinux/ott/q-amlogic-20210208-gtvs.xml  
$repo sync
```

Notice: Oversea customer use below command:

```
$cd~/<your-project-repo-dir>/  
$repo init -u ssh://git@openlinux2.amlogic.com/q-amlogic/platform/manifest.git -b q-amlogic  
-m openlinux/ott/q-amlogic-20210208-gtvs.xml  
$repo sync
```

Needed storage space: need about 200G storage space.

2.5. How to Enable FCC/PIP functional

```
diff --git a/ohm.mk b/ohm.mk
index 2e9d3b6..b01136f 100644
--- a/ohm.mk
+++ b/ohm.mk
@@ -52,8 +52,8 @@ endif

ifeq ($(PRODUCT_SUPPORT_DTVKIT),true)

-#PRODUCT_SUPPORT_DTVKIT_PIP := true
-#PRODUCT_SUPPORT_DTVKIT_FCC := true
+PRODUCT_SUPPORT_DTVKIT_PIP := true
+PRODUCT_SUPPORT_DTVKIT_FCC := true
```

Noticed: mxl258c supports FCC/PIP in S905X4 platform.

2.6. How to Build Code

Build Bootloader

1. ohm(S905X4)

```
cd bootloader/u-boot-repo/  
./mk sc2_ah212 --avb2  
cp build/u-boot.bin.signed ../../device/amlogic/ohm/bootloader.img  
cp build/u-boot.bin.sd.bin.signed ../../device/amlogic/ohm/upgrade/  
cp build/u-boot.bin.usb.signed ../../device/amlogic/ohm/upgrade/
```

Noticed: sc2_ah212 for ddr4.

sc2_ah212_ddr3 for ddr3.

sc2_ah212_lpddr3 for lpddr3.

sc2_ah212_lpddr4 for lpddr4.

2. franklin_hybrid(S905X2)

```
cd bootloader/u-boot-repo/  
./mk g12a_u215_v1 --bl32 ../../vendor/amlogic/common/tdk/secureos/g12a/bl32.img --avb2  
cp build/u-boot.bin ../../device/amlogic/franklin/franklin_hybrid/bootloader.img  
cp build/u-boot.bin.sd.bin ../../device/amlogic/franklin/franklin_hybrid/upgrade/u-boot.bin.sd.bin  
cp build/u-boot.bin.usb.bl2 ../../device/amlogic/franklin/franklin_hybrid/upgrade/u-boot.bin.usb.bl2  
cp build/u-boot.bin.usb.tpl ../../device/amlogic/franklin/franklin_hybrid/upgrade/u-boot.bin.usb.tpl
```

3. newton_hybrid(S905C1)

```
cd bootloader/u-boot-repo/  
./mk sm1_ac232_v1 --bl32 ../../vendor/amlogic/common/tdk/secureos/g12a/bl32.img --avb2  
cp build/u-boot.bin ../../device/amlogic/newton/newton_hybrid/bootloader.img  
cp build/u-boot.bin.sd.bin ../../device/amlogic/newton/newton_hybrid/upgrade/u-boot.bin.sd.bin  
cp build/u-boot.bin.usb.bl2 ../../device/amlogic/newton/newton_hybrid/upgrade/u-boot.bin.usb.bl2  
cp build/u-boot.bin.usb.tpl ../../device/amlogic/newton/newton_hybrid/upgrade/u-boot.bin.usb.tpl
```

Build Android

1. ohm(S905X4)

```
. build/envsetup.sh  
lunch ohm-user  
make -j8
```

2. franklin_hybrid(S905X2)

```
. build/envsetup.sh  
lunch franklin_hybrid-user  
make -j8
```

3. newton_hybrid(S905C1)

```
. build/envsetup.sh  
lunch newton_hybrid-user  
make -j8
```

Amlogic Hybrid android Q V20210208

2.7. Device Vendor SCS(Secure boot v3)

Device Vendor SCS(Secure boot v3), please refer the documents.

- <Amlogic S905X4 DeviceVendor SCS SigningProcess User Guide>
- <Amlogic S905X4 DIFPassword GenerationUser Guide>
- <Amlogic S905X4 USBPassword GenerationUser Guide>
- <Amlogic S905X4 Firmware Anti-rollback User Guide>
- <Amlogic S905X4 EFUSEBurning Provision UserGuide>

Device Vendor SCS Build Android.

```
copy u-boot.bin.device.signed to device/amlogic/ohm/bootloader_secure.img
```

```
enable PRODUCT_USE_PREBUILD_SECURE_BOOTLOADER := true in device/amlogic/ohm/ohm.mk
```

```
. build/envsetup.sh
```

```
lunch ohm-user
```

```
make -j8
```

S905X2 and S905C1 please refer the documents:

- <Aml_Linux_SecureBootV3_SignTool>

2.8. Secure Mode

This document provides recommended security mode configuration guidelines to prevent system data from being tampering with. Please refer the documents:

- <Security Mode Configuration Guide>

Amlogic Hybrid android Q V20210208

2.9. Fastboot

2.9.1.adb/fastboot version

We **MUST** update adb/fastboot version to 29.0.1-5644136 or 29.0.1-5644136+. Otherwise there are lots of adb/flash issues.

adb:

```
Android Debug Bridge version 1.0.41  
Version 29.0.1-5644136
```

fastboot

```
fastboot version 29.0.1-5644136
```

You can download latest SDK Platform-Tools from

<https://developer.android.com/studio/releases/platform-tools.html#downloads>

2.9.2. Two fastboot modes

Currently fastboot is split to bootloader space and fastbootd space(user space).

1. Bootloader mode

Bootloader space is similar as before.

It supported physical partitions: **bootloader**, **dts**, **boot**, **super**, **recovery**, etc

How to enter bootloader mode?

```
$ adb reboot bootloader
```

fastboot commands in bootloader mode

```
fastboot devices
fastboot reboot
fastboot reboot-bootloader
fastboot flashing unlock_critical
fastboot flashing unlock
fastboot flashing lock_critical
fastboot flashing lock
fastboot flashing get_unlock_ability

fastboot flash bootloader bootloader.img
fastboot flash bootloader-boot0 bootloader.img
fastboot flash bootloader-boot1 bootloader.img

fastboot flash dts dt.img
fastboot flash dtbo dtbo.img
fastboot flash vbmeta vbmeta.img
fastboot flash logo logo.img
fastboot flash boot boot.img
fastboot flash super super.img //not support burn logic partition in bootloader
fastboot flash recovery recovery.img

fastboot erase env/param/tee/boot/recovery/dtb
fastboot -w
fastboot format cache

fastboot getvar all
fastboot getvar is-userspace // return no in bootloader
fastboot getvar is-logical:boot //return no, others
fastboot getvar is-logical:system //return yes, system/odm/vendor/product
fastboot getvar super-partition-name //return super
```

2. Fastbootd mode

Fastbootd space is new.

It supported to flash all logical partitions, currently there are:system, vendor, product, odm

How to enter fastbootd mode?

```
$ adb reboot fastboot
```

fastboot commands in fastbootd mode

Amlogic Openlinux Release Notes

```
fastboot devices
fastboot reboot
fastboot reboot-bootloader
```

```
fastboot getvar is-userspace      // return yes in fastbootd
fastboot getvar is-logical:boot   //return no, others
fastboot getvar is-logical:system //return yes, system/odm/vendor/product
fastboot getvar super-partition-name //return super
fastboot getvar all
```

```
fastboot flash system system.img
fastboot flash odm odm.img
fastboot flash product product.img
fastboot flash vendor vendor.img
```

```
fastboot resize-logical-partition product 629145600 // odm/vendor/system/product
fastboot delete-logical-partition odm
fastboot create-logical-partition odm 134217728
```


2.9.3.How to remount

1. enter bootloader fastboot mode

```
fastboot flashing unlock  
fastboot reboot
```

2. run in adb mode

```
adb root  
adb disable-verity  
adb reboot  
#####adb wait-for-device#####  
adb root  
adb shell setenforce 0      # this cmd is needed by remount system & vendor, for product/odm, you can  
ignore it  
adb remount
```

Note: This method only works for the userdebug version.

2.10. How to Upgrade

There are 3 ways for update.

- Upgrade with USB burn tool
- Upgrade with OTA
- Upgrade with fastboot

2.10.1. all partitions

If you like to use our usb burn tool to update the images, it's the same as before. We have a fastboot flash-all script to update image, just like Android Q. You need not care about the dynamic partitions.

Partition name	make command	image name	burning ways
all partitions	make otapackage	1.ohm-ota-20210208.zip 2.aml_upgrade_package.img 3.ohn-fastboot-20210208.zip	1.by OTA upgrade 2.by USB burning 3.by fastboot
	make aml_upgrade	aml_upgrade_package.img	by USB burning

Amlogic Openlinux Release Notes

2.10.2. physical partitions

In Android Q, physical partitions is **bootloader, dts, boot, super, recovery, etc.**

Fastboot should be in **bootloader** mode when flash these physical partitions.

Below is the way of burning each partitions:

Partition name	make command	imgae name	burning ways
boot	make bootimage	boot.img	In uboot cmdline: U disk: usb_update boot boot.img SD card: sdc_update boot boot.img In bootloader mode: Fastboot: fastboot flash boot boot.img
logo	make logoimg	logo.img	usb_update logo logo.img sdcard_update logo logo.img Fastboot: fastboot flash logo logo.img
recovery	make recoveryimage	recovery.img	usb_update recovery recovery.img sdcard_update recovery recovery.img Fastboot: fastboot flash recovery recovery.img
super	make superimage	super.img	Fastboot: fastboot flash super super.img
dtb	make dtbimage	dt.img	usb_update _aml_dtb dt.img Fastboot: fastboot flash dts dt.img
uboot	./mk sc2_ah212 --avb2	u-boot.bin.signed	usb_update bootloader u-boot.bin sdcard_update bootloader u-boot.bin Fastboot: fastboot flash bootloader u-boot.bin

Notes: execute 'fastboot flashing unlock' to unlock the device if wants to use fastboot commands.

Amlogic Openlinux Release Notes

2.10.3. logical partitions

In Android Q, logical partitions is **system, vendor, odm, product**.

Fastboot should be in fastbootd mode when flash these logical partitions.

Below is the way of burning each partitions:

Partition name	make command	imgae name	burning ways
system	make systemimage	system.img	usb_update system system.img sdcard_update system system.img In fastbootd mode: Fastboot: fastboot flash system system.img
vendor	make vendorimage	vendor.img	usb_update vendor vendor.img sdcard_update vendor vendor.img In fastbootd mode: Fastboot: fastboot flash vendor vendor.img
odm	make odm_image	odm.img	usb_update odm odm.img sdcard_update odm odm.img In fastbootd mode: Fastboot: fastboot flash odm odm.img
product	Make productimage	product.img	In fastbootd mode: Fastboot: fastboot flash product product.img

Notes: execute 'fastboot flashing unlock' to unlock the device if wants to use fastboot commands.

3. Test Report

3.1. Auto Reboot/Shutdown in Normal Temperature

Test Platform	Test Case	Total test time	Test Result
AH212#175	Auto reboot	72H	Pass
AH212#194	Auto reboot	72H	Pass
AH212#195	Auto reboot	72H	Pass
AH212#200	Auto reboot	72H	Pass
AH212#195	Auto shutdown	72H	Pass
AH212#200	Auto reboot	72H	Pass
AH212#248	Auto reboot	72H	Pass
AH212#249	Auto reboot	72H	Pass

3.2. Auto Suspend in Normal Temperature

Test Platform	Test Case	Total test time	Test Result
AH212#101	Auto suspend	72H	Pass
AH212#105	Auto suspend	72H	Pass
AH212#162	Auto suspend	72H	Pass
AH212#163	Auto suspend	72H	Pass
AH212#195	Auto suspend	72H	Pass
AH212#200	Auto suspend	72H	Pass
AH212#248	Auto suspend	72H	Pass
AH212#249	Auto suspend	72H	Pass

3.3. 3D Game burning Test Report

Test Platform	Test Case	Total test time	Test Result
AH212#1	3D Game	72H	Pass
AH212#2	3D Game	72H	Pass
AH212#3	3D Game	72H	Pass
AH212#4	3D Game	72H	Pass
AH212#5	3D Game	72H	Pass
AH212#176	3D Game	72H	Pass
AH212#186	3D Game	72H	Pass
AH212#237	3D Game	72H	Pass
AH212#238	3D Game	72H	Pass

3.4. Monkey test Test Report

Test Platform	Test Case	Total test time	Test Result
AH212#175	Monkey test/300ms	72H	Pass
AH212#194	Monkey test/300ms	72H	Pass
AH212#295	Monkey test/300ms	72H	Pass
AH212#200	Monkey test/300ms	72H	Pass

Monkey Black list

Amlogic Openlinux Release Notes

com.droidlogic.PPPoE
com.droidlogic.miracast
com.droidlogic.mediacenter
com.droidlogic.otaupgrade
com.droidlogic.tvinput
com.hpplay.happyplay.aw
com.android.tv.settings
com.droidlogic.tv.settings
com.google.android.tungsten.setupwraith
com.google.android.gms
com.android.vending
com.google.android.youtube.tv
com.google.android.videos
com.google.android.tv.remote.service
com.google.android.tvrecommendations
com.google.android.gms.persistent
com.google.android.youtube.tv.recommendations
com.google.android.tv.frameworkpackagestubs
com.google.android.tvcom.droidlogic.videoplayer

3.5. FactoryReset Test Report

Test Platform	Test Case	Total test time	Test Result
AH212#175	Factory reset	72H	Pass
AH212#194	Factory reset	72H	Pass
AH212#295	Factory reset	72H	Pass
AH212#200	Factory reset	72H	Pass

3.6. A/V Burn Test Report

Test Platform	Test Case	Total test time	Test Result	Comments
AH212#176	4K AV1	72H	Pass	
AH212#186	4K AV1	72H	Pass	
AH212#237	4K H264	72H	Pass	
AH212#238	4K H264	65H	Fail	Exit player
AH212#248	4K H265	72H	Pass	
AH212#249	4K H265	72H	Pass	
AH212#200	4K VP9	72H	Pass	
AH212#249	4K VP9	65H	Fail	Exit player
AH212#117	Youtube	12H	Pass	
AH212#120	Youtube	12H	Pass	

3.7. DVB Burn Test Report

Test Platform	Test Case	Total test time	Test Result	Comments
---------------	-----------	-----------------	-------------	----------

Amlogic Openlinux Release Notes

AH212#123	Stream playback	72H	Pass	
AH212#124	Stream playback	72H	Pass	
AH212#125	Stream playback	72H	Pass	
AH212#126	Stream playback	72H	Pass	
AH212#123	Auto Search	12H	Pass	
AH212#124	Auto Search	12H	Pass	
AH212#125	Auto Search	12H	Pass	
AH212#126	Auto Search	12H	Pass	
AH212#123	Audio Switch	12H	Pass	
AH212#124	Audio Switch	12H	Pass	
AH212#125	Audio Switch	12H	Pass	
AH212#126	Audio Switch	12H	Pass	
AH212#123	PVR	12H	Pass	
AH212#124	PVR	12H	Pass	
AH212#125	PVR	12H	Pass	
AH212#126	PVR	12H	Pass	
AH212#1	Program Switch(PIP)	12H	Pass	
AH212#2	Program Switch(PIP)	12H	Fail	Tsplayer crash
AH212#1	Timeshift(PIP)	12H	Pass	
AH212#2	Timeshift(PIP)	12H	Fail	Can't show subtitle

3.8. High(70℃) Temperature Test Report

Test Platform	Test Case	Total test time	Test Result
AH212#1	Auto reboot	72H	Pass
AH212#2	Auto reboot	72H	Pass
AH212#3	Auto reboot	72H	Pass
AH212#4	Auto reboot	72H	Pass
AH212#5	Auto reboot	72H	Pass
AH212#1	3D Game	72H	Pass
AH212#2	3D Game	72H	Pass
AH212#3	3D Game	72H	Pass
AH212#4	3D Game	72H	Pass
AH212#5	3D Game	72H	Pass

3.9. Low(-20℃) Temperature Test Report

Test Platform	Test Case	Total test time	Test Result
AH212#1	Auto reboot	72H	Pass
AH212#2	Auto reboot	72H	Pass
AH212#3	Auto reboot	72H	Pass
AH212#4	Auto reboot	72H	Pass
AH212#5	Auto reboot	72H	Pass
AH212#1	3D Game	72H	Pass
AH212#2	3D Game	72H	Pass
AH212#3	3D Game	72H	Pass
AH212#4	3D Game	72H	Pass
AH212#5	3D Game	72H	Pass

3.10. Video Format Test Report

Extension	Codec Detail	Tested Resolution	Ohm
.3g2	H263	320x276	Support
.3gp	MPEG-4 Visual	640x480	Support
	MPEG-4	320x240	Support
	H263	320x276	Support
	H264	1920x1080	Support
.avi	AVC	1920x1080	Support
	M-JPEG	1024x576	Support
	RealMagic	720x480	Support
	MPEG-4	720x576	Support
	h264	1920x1080	Support
	FF mpeg MPEG4	640x480	Support
.f4v	AVC(H264)	1280x720	Support
.flv	Sorenson Spark	1920x1080	Support
	VP6	800x342	Support
	AVC(H264)	1920x1080	Support
.mp4	HEVC(H265)	1920x1080	Support
	4K HEVC(4K H265)	4096x2304	Support
	8K HEVC(8K H265)	8192x4320	No support
	AV1	1920x1080	Support
	4K AV1	3840x2160	Support
	H263	176x144	Support
	AVC	1920x1080	Support
.m2ts	VC-1	1920x1080	Support
	MPEG-2	480x576	Support
.m2v	AVC	1280x720	Support
.mkv	WMV3	1280x720	Support
	MPEG-4 Visual	1920x1080	Support
	AVC	1920x1080	Support
	4K HEVC(4K H265)	3840x2160	Support
	8K HEVC(8K H265)	8192x4320	No support
	VP8	1920x1080	Support
	VP9	1920x1080	Support
.mov	MPEG-4 Visual	1280x720	Support
	mjpa	640x480	Support
	H263	320x240	Support
	M-JPEG	640x480	Support
	AVC(H264)	1920x1080	Support
	4K H264	3840x2160	Support
.mpeg	MPEG-2	1920x1080	Support
	VC-1	1920x1080	Support
	MPEG-1	720x576	Support

Amlogic Openlinux Release Notes

.mts	AVC	1440x1080	Support
.ogm	XVID	640x480	Support
.PMP	H264	480x272	Support
.rmvb	RealVideo1	1280x720	Support
	RealVideo2	1280x720	Support
	RealVideo3	1280x720	Support
	RealVideo4	1920x1080	Support
.tp	MPEG-2	1920x1088	Support
.ts	HEVC(H265)	1920x1080	Support
	4K HEVC(4K H265)	4096x2304	Support
	8K HEVC(8K H265)	8192x4320	No support
	MPEG-1	1920x1080	Support
	avs	720x576	Support
	avs+	1920x1080	Support
	avs2	3840x2160	Support
	AVC	1920x1080	Support
.vob	MPEG-2	1920x1080	Support
	MPEG-2	720x576	Support
.wmv	WMV1	640x480	Support
	WMV2	768x432	Support
	WMV3	1920x1080	Support
.webm	VP8	1920x1080	Support
	VP9	1920x1080	Support
	4K VP9	4096x2160	Support
	8K VP9	8192x4608	No support
	AV1	1920x1080	Support
4K	AVC(H264)	4096x2304	Support
	HEVC(H265)	4096x2304	Support
	VP9	4096x2160	Support
	AV1	3840x2160	Support
8K	HEVC(H265)	8192x4320	No support
	VP9	8192x4608	No support

3.11. Power Consumption Test Report

Please contact with your Amlogic Sales/Technical support window for latest power consumption test report.

Amlogic Openlinux Release Notes

4. Tools version

Tool name	version
USB burning tool	Aml_Burn_Tool_V3.1.8
Customization	Amlogic customization V.2.1.0
Kernel tool chain	gcc-linaro-6.3.1-2017.02-x86_64_arm-linux-gnueabi gcc-linaro-6.3.1-2017.02-x86_64_aarch64-linux-gnu

5. License version

Licensed Feature Name	Version	Document version
Dolby Digital Plus for Android	v20201021	In firmware Readme text
Dolby Digital for Android	v20200723	In firmware Readme text
Dolby MS12 V 1.3.2 for Android	v20201215	In firmware Readme text
DTS-HD for android	v20200715	In firmware Readme text
DRA Decoder for android	v20200522	In firmware Readme text
Secure OS TDK	Tdk is included in the code	TDK For SC2-S905X4 Android Q only Amlogic TDK Integration User Guide V3.0 Secure Key Provision For Android Q Amlogic Secure Key Provision User Guide V2.4
Widevine	L1/L3	Amlogic WideVine Integration User Guide v0.9
Playready	openlinux/ott/q-amlogic (189ad8d8976f465cbdc35a81b11 2c7c539506204)	Amlogic PlayReady Integrate User Guide v1.32
Dolby Vision 2.5 for Android kernel64bit	For SC2-S905X4 stb_release_20201005	In firmware Readme text
HDCP 1.4 for HDMI TX for Android	v20180408	Amlogic_HDCP1.4_Tx_Release_Note_V0.1
HDCP 2.X for HDMI TX for Android	v20201212	Amlogic_HDCP2.2_Tx_Release_Note_V1.0
DTVKit		dtvkit Integrate User Guide v0.3
Nagra	nagra/nagra-cashal-rel(master: 53837a1750d97bd28b60c793666 b503c0646f251) nagra/nagra-sdk(master: 6748831104a41467212cd0afbe4e e8fcbcbce6832)	Amlogic CAS - Nagra SDK Integration User Guide v0.5
Verimatrix	vmx_release_binaries(master: c0281799b228c8852173756eb68 9b8023afff2f1) vendor/vmx/bootloader(master: 0d1974225f0b8c5db2e1ea52a6f3 5735bf37a2fb) vendor/amlogic/prebuild/verimatrix (q-amlogic:	Amlogic CAS - Verimatrix SDK Integration User Guide v0.4 Amlogic CAS - Verimatrix Client Integration User Guide-v3.0

Amlogic Openlinux Release Notes

	9a1ca77d490f30ba8c84813c94c367fd006cf53f) vmx/sdk-rel(m9x4-rel: 9a6dd580e8ce8039fd013e46aa06a677a86ed69a)	
irdeto	irdeto-imw (efa9fc52ac9a3992c1ab4898405b3a8f768a2f1a) irdeto-sdk(d8593ce198ffcdea4d732ee7c26f016826fad1d3)	Amlogic CAS - Irdeto SDK Integration User Guide v1.5.docx Amlogic S905C1 Irdeto ODM Code Signing Guide v1.9 S905C1 Series MSR3 Secure Chipset Configuration Guide v2.3 Amlogic Irdeto-IMW-Integration-Guide-v20200519.doc

6. Licence path

Licensed Files	
HDCP TX firmware.le	/odm/etc/firmware/firmware.le
Dolby	/odm/lib/libHwAudio_dcvdec.so
MS12	/odm/lib/libdolbys12.so
Dolby vision	/odm/lib/modules/dovi.ko

7. Attention

In the platform of S905X2/S905C1, the baud rate is 115200 and the command to enter usb burn mode is update.

In the platform of S905X4, the baud rate is 921600 and the command to enter usb burn mode is adnl.